

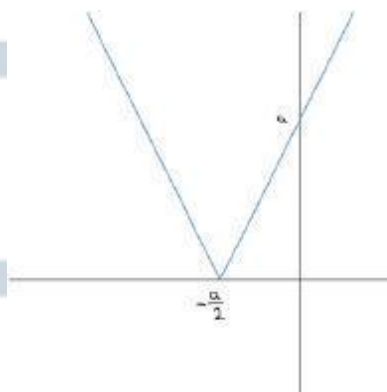
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dx}{dt}$, t , x and a constant of proportionality.	AO3.3	B1	$\frac{dx}{dt} = \frac{k(8-t)}{x}$
	Substitutes $t = 2$, $x = 336$, $\frac{dx}{dt} = 72$ to find k	AO1.1a	M1	$72 = \frac{k(8-2)}{336}$ $k = 4032$
	Obtains correct value of k shows given result AG	AO2.1	R1	$\frac{dx}{dt} = \frac{4032(8-t)}{x}$ $x \frac{dx}{dt} = 4032(8-t)$
(b)	Integrates one side correctly	AO1.1a	M1	$\int x dx = \int 4032(8-t) dt$
	Integrates both sides correctly condone missing c	AO1.1b	A1	$\frac{1}{2}x^2 = 4032\left(8t - \frac{t^2}{2}\right) + c$
	Uses conditions to show $c = 0$ and correctly obtains given result. AG	AO2.1	R1	$\frac{1}{2} \times 336^2 = 4032\left(8 \times 2 - \frac{2^2}{2}\right) + c$ $c = 0$ $x^2 = 64512t - 4032t^2$ $\therefore x^2 = 4032t(16-t)$
(c) (i)	Translates rate of growth into $\frac{dx}{dt} = 24$ and uses in model for rate of sales	AO3.3	M1	$24x = 4032(8-t)$ $x = 168(8-t)$ $(168(8-t))^2 = 4032t(16-t)$
	Eliminates x to form quadratic equation or inequality in t	AO3.1a	M1	$t^2 - 16t + 56 = 0$ $t = 5.171\ldots$ or $10.828\ldots$ 5 hours 10 minutes
	Obtains correct equation or inequality in t Need not be simplified	AO1.1b	A1	Earliest time 14:40
	Obtains $t = 5.17$	AO1.1b	A1	
	Converts their t into hours and minutes	AO3.4	A1	

	Interprets the closing time as 14:40 09:30 + their converted time	AO3.2a	R1F	
(ii)	Explains in context that when the stall opens there will be zero sales	AO3.5a	E1	When the stall opens there are zero sales
	Explains that when $x = 0$ the model is undefined	AO3.5a	E1	When $x = 0$ $\frac{dx}{dt}$ is undefined as the denominator is zero
Total 14 marks				

Q2.

Marking Instructions	AO	Marks	Typical Solution
Draws the correct V-shape, nothing below x -axis	AO1.2	M1	
Intersects negative x -axis with $-\frac{a}{2}$ labelled	AO1.1b	A1	
Intersects positive y -axis with a labelled	AO1.1b	A1	
			Total 3 marks

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Q3.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$ x < \frac{3}{2}$
Total 1 mark			

Q4.

- (a) (i) $p(3) = 3^3 - 4 \times 3^2 - 3 \times 3 + 18$
 $p(3)$ attempted **not** long division

M1

$p(3) = 27 - 36 - 9 + 18 = 0 \Rightarrow x - 3$ is a factor
shown = 0 plus statement

A1

2

- (ii) Quadratic factor ($x^2 - x + c$) or ($x^2 + bx - 6$)
-x or -6 term by inspection
or full long division by $x - 3$
or comparing coefficients
or $p(-2)$ attempted

M1

Quadratic factor ($x^2 - x - 6$)
correct quadratic factor (or $x + 2$ shown to be factor by Factor Theorem)

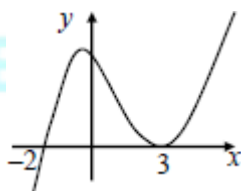
A1

$[p(x) =] (x - 3)(x - 3)(x + 2)$
or $[p(x) =] (x - 3)^2(x + 2)$
must see product of factors

A1

3

(b)



cubic curve with one maximum and one minimum

M1

meeting x-axis at -2 and touching x-axis at 3

A1

Final A1 is dependent on previous A1 and can be withheld if curve
 has very poor curvature beyond $x = 3$, V shape at $x = 3$ etc

*graph as shown, going beyond $x = -2$ but condone
 max on or to right of y -axis*

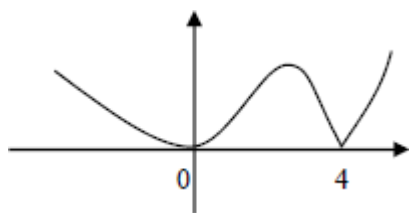
A1

3

[8]

Q5.

(a)



reflection in the x -axis for the negative $f(x)$ and remainder as given on sketch

M1

*correct curvatures, correct cusp at $x = 4$
condone straight lines for $x < 4$
 4 marked on x -axis*

A1

2

(b) **Either**

1. Stretch

1 and either 2 or 3

M1

2. $\parallel x$ -axis

3. by factor 0.5

1, 2 and 3

A1

(followed by) translation

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$$\begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$$

B1

4

or
translation

(E1)

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(B1)

(followed by) 1. Stretch
1 and either 2 or 3

(M1)

2. || x-axis

3. by factor 0.5
1, 2 and 3

(A1)

(4)

[6]

Q6.

(a) $(2x - 3 = x)$

$x = 3$

B1

$2x - 3 = -x$

or $-(2x - 3) = x$ or $-2x + 3 = x$

M1

$x = 1$

A1

3

(b) $(|2x - 3| \geq |x|)$

No ISW in part(b), mark their final line as their answer.

$x \leq 1$

Or $1 \geq x$

B1

$x \geq 3$

Or $3 \leq x$

Or " $x \leq 1$ or $x \geq 3$ " for B1 B1

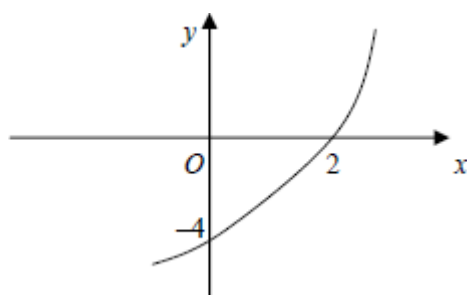
B1

2

[5]

Q7.

(a)



Reflection in the x -axis.

M1

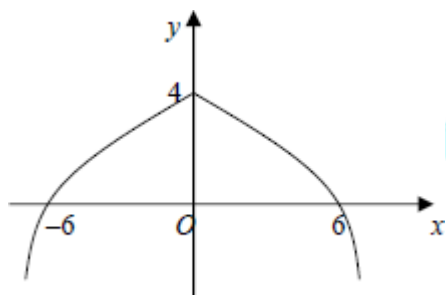
Intersection with the x -axis and y -axis marked 2 and -4 .

Accept $(2, 0)$ and $(0, -4)$ instead of marking on the axes.

A1

2

(b)



Reflection of $0 < x < 6$ part in the y -axis giving two connected sections

M1

Correct curve beyond ± 6 , correct curvature and correct cusp at $x = 0$ (generous)

A1

± 6 and 4 marked correctly

Accept $(6, 0)$, $(0, 4)$ and $(-6, 0)$ instead of marking on the axes.

A1

3

(c) Reflection

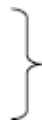
M1

in the y -axis

A1

(followed by)

- 1) stretch
 2) parallel to the x -axis
 3) by factor 2



1 and either 2 or 3

M1

1, 2 and 3

OR

Vice versa

A1

4

[9]

Q8.

- (a) (i) $p(-1) = (-1)^3 + 2(-1)^2 - 5(-1) - 6$
 *$p(-1)$ attempted **not** long division*

M1

$p(-1) = -1 + 2 + 5 - 6 = 0 \Rightarrow x + 1$ is a factor
CSO; correctly shown = 0 plus statement

A1

2

- (ii) Quad factor in this form: $(x^2 + bx + c)$
long division as far as constant term
or comparing coefficients,
or $b = 1$ or $c = -6$ by inspection

M1

$$x^2 + x - 6$$

correct quadratic factor

A1

$[p(x) =] (x + 1)(x + 3)(x - 2)$
***must** see correct product*

A1

3

- (b) $p(0) = -6$; $p(1) = -8$
***both** $p(0)$ and $p(1)$ attempted*
***and** at least one value correct*

M1

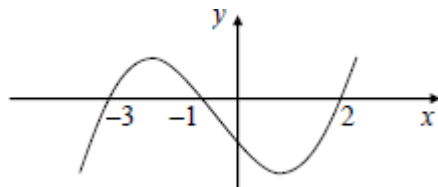
$$\Rightarrow p(0) > p(1)$$

AG both values correct plus correct statement involving $p(0)$ and $p(1)$

A1

2

(c)



cubic with one max and one min

M1

$\sim$ with $-3, -1, 2$ marked

A1

correct with minimum to right of y -axis AND going beyond -3 and 2

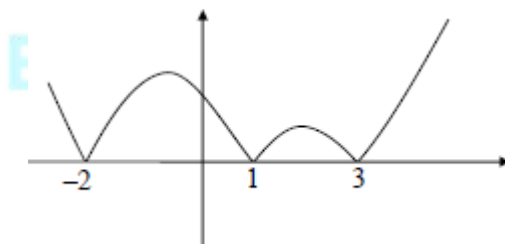
A1

3

[10]

Q9.

(a)



Modulus graph, 4 sections touching x -axis at $-2, 1, 3$

M1

Correct $x > 3, x < -2$

A1

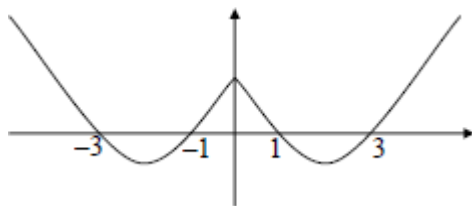
Correct $-2 \leq x \leq 3$ with maximum at 2 lower than maximum at -1 and correct cusps at $x = -2, x = 1$ and $x = 3$

The maximums need to be at $x = -1$ and 2 (approx)

A1

3

(b)



Symmetrical about y -axis, from original curve for
 $0 < x < 1$ and $x > 3$

M1

Correct graph including cusp at $x = 0$

A1

2

(c)

Translate $\left\{ \begin{matrix} -1 \\ 0 \end{matrix} \right\}$
 Stretch (I) $\left\{ \right\}$
 sf $\frac{1}{2}$ (II) $\left\{ \right\}$
 // y -axis (III) $\left\{ \right\}$ either order

I and (either II or III)

E1
 B1

I + II + III

M1

A1

4

(d) $x = -2$

B1

$y = 5$

Each value may be stated or shown as coordinates

B1

2

[11]

Q10.

(a) $-3 \leq f(x) \leq 3$

$$\begin{aligned}
 -3 \leq x \leq 3, -3 < f(x) < 3 \\
 -3 < f < 3, -3 < y < 3 \\
 -3 \leq f < 3, -3 < f \leq 3
 \end{aligned}$$

M1

$$\text{Allow } -3 \leq y \leq 3, -3 \leq f \leq 3$$

A1

2

(b) (i) $y = 3 \cos \frac{1}{2}x$

$$\frac{y}{3} = \cos \frac{1}{2}x$$

$$\cos^{-1} \frac{y}{3} = \left(\frac{1}{2}x \right)$$

$$x = 2 \cos^{-1} \frac{y}{3}$$

$$y = 2 \cos^{-1} \frac{x}{3}$$

$$\left. \begin{array}{l} \text{Or } \cos^{-1} \frac{x}{3} = \\ \text{Swap } x \text{ and } y \end{array} \right\} \text{Either order}$$

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M1

M1

$$f^{-1}(x) = 2 \cos^{-1} \frac{x}{3}$$

A1

3

(ii) $\frac{x}{3} = \cos \frac{1}{2}$

*If incorrect in (b)(i) **BUT** answer in form $p \cos^{-1}(qx)$ (condone $p, q = 1$)*

$$\text{Then } qx = \cos\left(\frac{1}{p}\right) \text{ M1 or } x = f\left(\frac{1}{p}\right) \text{ M1}$$

M1

$$x = 3 \cos \frac{1}{2} \quad \text{ISW}$$

$$x = 3 \cos \frac{1}{2} \quad \text{A1}$$

A1

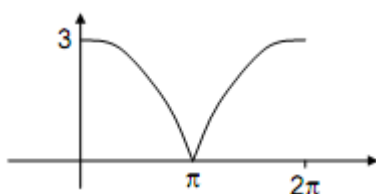
2

(c) (i) $gf(x) = \left| 3 \cos \frac{1}{2} x \right|$

B1

1

(ii)



Modulus graph in 1st quadrant, starting from a +ve y-intercept, at least 2 continuous parts, first descending, then second increasing
 IGNORE CURVE OUTSIDE RANGE

M1

Correct curvature, curves reaching x-axis, condone multiple curves (no turning points at axis)

A1

Approximately symmetrical graph with 3, π , 2π indicated (must have scored previous 2 marks)

Condone $y = 3 \cos \frac{1}{2} x$ also drawn but clearly identified, otherwise M0

A1

3

(d) STRETCH + direction

Either in x-direction or y-direction

M1

s.f. 3, parallel to y-axis

A1

s.f. 2, parallel to x -axis
Either order

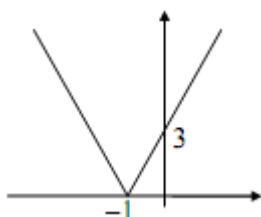
A1

3

[14]

Q11.

(a) (i)



*modulus graph, approximate V shape,
 touching negative x -axis and crossing y -axis*

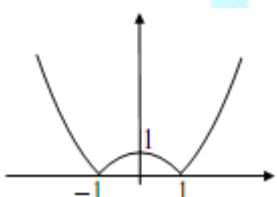
M1

-1, 3 marked, graph symmetrical, straight lines

A1

2

(ii)



*modulus graph in 3 sections, touching
 x -axis and crossing positive y -axis*

M1

$$\left. \begin{array}{l} \text{correct curvature} \\ \text{their } x > 1, \text{ their } x < -1 \\ \text{correct curve } -1 \leq x \leq 1 \\ \text{and } x = \pm 1, y = 1 \text{ marked} \end{array} \right\} \text{independent}$$

A1

A1

3

(b) (i) $|3x + 3| = |x^2 - 1|$

$(3x + 3 = x^2 - 1)$

$(0 \Rightarrow) x^2 - 3x - 4 \quad \text{---A}$

either A or B seen, all terms on one side

M1

$$x = 4, -1$$

A1,A1

$$(3x + 3 = 1 - x^2)$$

$$x^2 + 3x + 2 (= 0) \quad \text{---B}$$

$$x = -1, -2$$

A1,A1

5

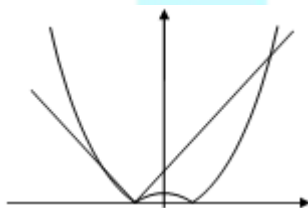
$$\therefore x = -2, -1, 4$$

SC NMS or partial method

$\left. \begin{array}{l} 1 \text{ correct value } 1/5 \\ 2 \text{ correct values } 2/5 \\ 3 \text{ correct values } 3/5 \end{array} \right\} \text{ independent of method mark}$

more than 3 distinct values max 2/5

(ii)



$$x > 4, x < -2$$

$x >$ their largest, $x <$ their smallest;

CAO

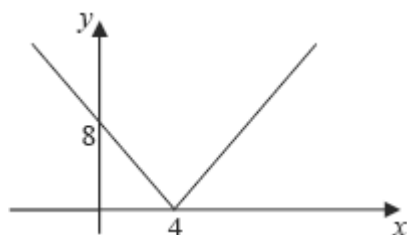
M1,A1

2

[12]

Q12.

(a)



Modulus graph V shape in 1st quad going into 2nd quad, touching x-axis. Must cross y-axis
Condone not ruled

M1

4 and 8 labelled

A1

2

(b) $x = 2$

One correct answer

B1

$x = 6$

Second correct answer and no extras
Condone answers shown on the graph, if clearly indicated

B1

2

(c) $x > 6$

One correct answer

B1

$x < 2$

Second correct answer and no extras and no further incorrect statement eg

$6 < x < 2$ or $2 < x > 6$

SC $x \geq 6, x \leq 2$ scores B1

B1

2

[6]

Q13.

(a) (i) $p(-3) = (-3)^3 + 7(-3)^2 + 7(-3) - 15$
 $p(-3)$ attempted; NOT long division

$= -27 + 63 - 21 - 15$

This line alone implies M1

M1

$p(-3) = 0 \Rightarrow (x + 3)$ is factor

$p(-3)$ shown = 0 plus statement

A1

2

(ii) $p(x) = (x + 3)(x^2 + px + q)$

*Full long division, comparing coefficients
or by inspection either $p = 4$ or $q = -5$*

M1

(Quadratic factor) $(x^2 + 4x - 5)$

*or M1A1 for either $x - 1$ or $x + 5$
clearly found using Factor Theorem*

A1

$(p(x) =) (x + 3)(x - 1)(x + 5)$

*Must be seen as a product of 3 factors
NMS full marks for correct product*

SC B2 for 3 correct factors listed NMS

SC B1 for $(x + 3)(x - 1)()$

or $(x + 3)(x + 5)()$

or $(x + 3)(x + 1)(x - 5)$

A1

3

(b) (i) $p(-1) = -16$; $p(0) = -15$

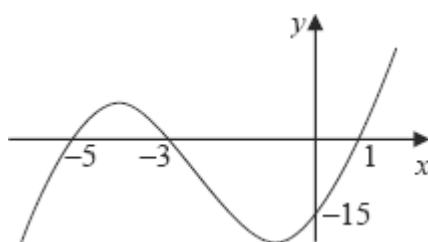
$\Rightarrow p(-1) < p(0)$

Values must be evaluated correctly

B1

1

(ii)



y-intercept -15 marked or $(0, -15)$ stated

B1

Cubic graph – 1 max, 1 min

M1

~ shape with -5, -3, 1 marked

A1

Graph correct with minimum point to left

of y-axis and going beyond both -5 and 1
Previous A1 must be scored

A1

Cannot score M1A0A1 but can score B0M1A1A1

4

[10]

Q14.

(a) $\int \frac{dx}{\sqrt{x+1}} = \int -\frac{1}{5} dx$

Correct separation; or $\frac{dt}{dx} = -5(x+1)^{\frac{1}{2}}$
Condone missing integral signs

B1

$$2\sqrt{x+1} = -\frac{1}{5}t \quad (+C)$$

Correct integrals; condone

$$\frac{\sqrt{x+1}}{\frac{1}{2}}$$

B1B1

$$x = 80 \quad t = 0 \quad C = 2\sqrt{81}$$

Use (0, 80) to find a constant C

M1

$$= 18$$

F on integrals if in form $\sqrt{x+1} = qt + c$

A1F

$$x = \left(9 - \frac{1}{10}t\right)^2 - 1$$

OE; CSO; x = correct expression in t

A1

6

(b) $t = 60 \quad x = f(60)$

Evaluate $f(60)$, ie $x = \dots$ (C not required)

M1

= 8

CSO

A1

2

(c) (i) $\frac{dA}{dt} = kA(9 - A)$

$\frac{dA}{dt} = \text{product involving } A; k \text{ required}$

Condone terms in t

M1A1

2

(ii) $4.5 = \frac{9}{1 + 4e^{-0.09t}}$

Condone one slip in denominator

M1

$e^{-0.09t} = \frac{1}{4}$

A1

$-0.09t = \ln\left(\frac{1}{4}\right)$

Take $\ln$ correctly

m1

$t = \frac{\ln\left(\frac{1}{4}\right)}{-0.09}$

= 15.4 (hours)

CAO; condone more than 3sf if correct
15.40327068 Allow 15h 24m

A1

4

[14]

Q15.

(a) (i) $p(-2) = -8 + 2 + 6$

$p(-2)$ attempted : **NOT** long division

M1

$p(-2) = 0 \Rightarrow x + 2$ is factor

Shown = 0 plus statement

May make statement first such as " $x + 2$ is a factor if $p(-2) = 0$ "

A1

Minimum statement required "factor"

2

(ii) $b = -2$

No working required for B1 + B1

B1

$c = 3$

Try to mark first using B marks

B1

or long division/comparing coefficients

Award M1 if B0 earned and a clear method is used

(M1)

$p(x) = (x + 2)(x^2 - 2x + 3)$

Must write final answer in this form if long division has been used to get A1

(A1)

2

(iii) $b^2 - 4ac = (-2)^2 - 4 \times 3$

Discriminant correct from their quadratic

M0 if $b = -1$, $c = 6$ used (using cubic eqn)

M1

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$b^2 - 4ac = -8$ (or < 0) $\Rightarrow$ no (other) real roots

CSO All values must be correct plus statement

A1

Or $(x - 1)^2 + 2$

Completion of square for their quadratic

(M1)

$(x - 1)^2 + 2 > 0$ therefore no real roots

Shown to be positive plus statement regarding no real roots

(A1)

Or $(x - 1)^2 = -2$ has no real roots

- (b) (i) ($y_B =$) 6
 Condone (0, 6)

B1

1

(ii) $\frac{x^4}{4} - \frac{x^2}{2} + 6x$

One term correct

M1

Another term correct

A1

All correct (ignore + c or limits)

A1

$$\left[\right]_{-2}^0 = 0 - (4 - 2 - 12)$$

$$= 10$$

$F(-2)$ attempted

m1

= 10

CSO Clearly from $F(0) - F(-2)$

A1

5

(iii) Area of $\Delta = \frac{1}{2} \times 2 \times 6$

Condone – 2 and ft their y_B value

or $\int_{-2}^0 (3x + 6) dx$ and attempt to integrate

M1

= 6

Must be positive allow –6 converted to +6

A1

Shaded region area = $10 - 6 = 4$

CSO 10 must come from correct working

A1

Q16.

(a) (i) $p(-1) = -1 + 7 - 6$
Finding $p(-1)$

M1

$= 0$ therefore $x + 1$ is a factor
*Shown to = 0 **plus statement***

A1

2

(ii) $p(x) = (x + 1)(x^2 - x - 6)$
Long division/inspection (2 terms correct)

M1

Quadratic factor correct

A1

$p(x) = (x + 1)(x + 2)(x - 3)$
*May earn M1, A1 For correct second factor
 then A1 For $(x + 1)(x + 2)(x - 3)$*

A1

3

(b) (i) $A(-2, 0)$
Condone $x = -2$

B1

1

(ii) $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$
One term correct

M1

Another term correct

A1

(may have $+ c$ or not)
All correct unsimplified

A1

$$\left[\frac{81}{4} - \frac{63}{2} - 18 \right] - \left[\frac{1}{4} - \frac{7}{2} + 6 \right]$$

$F(3) - F(-1)$ attempted in correct order

m1

$$= -32$$

CSO; OE

A1

5

(iii) Area of shaded region = 32

FT their (b)(ii) but positive value needed

B1ft

1

(iv) $\frac{dy}{dx} = 3x^2 - 7$

One term correct

M1

All correct (no + c etc)

A1

When $x = -1$, gradient = -4

CSO

A1

3

(v) Gradient of normal = $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg $4y - x = 1$

A1

3

[18]

Q17.

- (a) (i) $p(-2) = -8 + 4 + 16 - 12$
NOT long division

M1

$= 0 \Rightarrow (x + 2)$ is factor
 *$p(-2)$ shown = 0 **and** statement*

A1

2

- (ii) Quad factor by comparing coefficients or
 $(x^2 + kx \pm 6)$ by inspection
Or full long division or attempt at Factor Theorem using $f(\pm 3)$

M1

$p(x) = (x + 2)(x^2 - x - 6)$

Correct quadratic factor or $(x - 3)$ shown to be factor by Factor Theorem

A1

$p(x) = (x + 2)^2(x - 3)$ or $(x + 2)(x + 2)(x - 3)$

*CSO; SC: B1 For $(x + 2)(x^{***})(x - 3)$ by inspection or without working*

A1

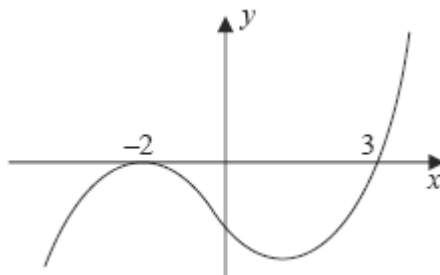
3

- (b) (i) $(k =) -12$
Condone $y = -12$ or $(0, -12)$

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1

(ii)



Cubic shape (one max and one min)

M1

*Maximum at $(-2, 0)$ and through $(3, 0)$ –
 at least one of these values marked*

A1

“correct” graph as shown
 (touching smoothly at -2 , 3 marked and
 minimum to right of y -axis)

A1

3

[9]

Q18.

(a) (i) $p \frac{dx}{dt} = q$

where p and q are functions

M1

$$\frac{dx}{dt} = -kx$$

in any correct combination

A1

2

(ii) $-500 = -k \ 20000$ or $500 = k \ 20000$
 condone sign error or missing 0
 k can be on either side of the equation

M1

$$k = \frac{5}{500} (= 0.025)$$

CSO both (a)(i) and (a)(ii)

A1

2

(b) (i) $A = 1300$

B1

1

(ii) $100 > Ae^{-0.05t}$

condone = for $>$; condone 99 for 100

M1

$$\ln \left(\frac{100}{A} \right) > -0.05t$$

take logs correctly

condone 0.5

m1

$$t > 51.3$$

*or by trial and improvement
(see end of mark scheme)*

A1

population first exceeds 1900 in 2059

F if M1 m1 earned and $t > 0$ following A

A1F

4

ALTERNATIVE METHOD

$$t = 51 \rightarrow 101.5$$

$t = 51$ or $t = 52$ considered

M1

$$t = 52 \rightarrow 96.6$$

$$\Rightarrow 51 < t < 52$$

population first exceeds 1900 in 2059

CAO

A3

4

[9]

Q19.

(a) (i) $p(-2) = -8 - 16 + 14 + k$

$$p(-2) = 0 \Rightarrow -10 + k = 0 \Rightarrow k = 10$$

Must have statement if $k = 10$ substitute

or long division or $(x + 2)(x^2 - 6x + 5)$

M1

AG likely withhold if $p(-2) = 0$ not seen

A1

2

(ii) $p(x) = (x + 2)(x^2 + \dots 5)$

$$p(x) = (x + 2)(x^2 - 6x + 5)$$

$$\Rightarrow p(x) = (x + 2)(x - 1)(x - 5)$$

Attempt at quadratic or second linear factor

M1

$(x - 1)$ or $(x - 5)$ from factor theorem

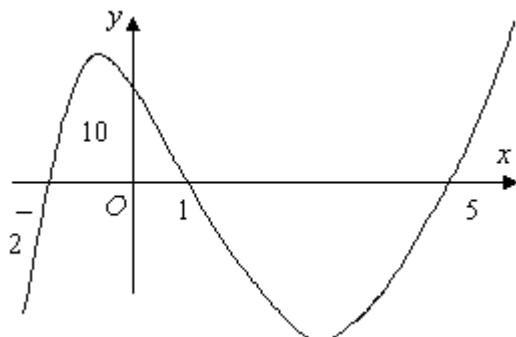
A1

Must be written as product

A1

3

(b)



Curve thro' 10 marked on y-axis

B1

FT their 3 roots marked on x-axis

B1 ✓

Cubic shape with a max and min

M1

*Correct graph (roughly as on left) going beyond -2 and 5
(condone max anywhere between $x = -2$ and 1 and min between 1 and 5)*

A1

4

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[9]

Q20.

(a) (i) $B(0, 5)$

B1

Area AOB $\frac{1}{2} \times 1 \times 5$

Condone slip in number or a minus sign

M1

$$= 2\frac{1}{2}$$

A1

3

(ii) $\frac{3x^6}{6} + \frac{2x^2}{2} + 5x$ or $\frac{x^6}{2} + x^2 + 5x$ (may have + c or not)

Raise one power by 1

M1

One term correct

A1

All correct unsimplified

A1

3

(iii) Area under curve = $\int_{-1}^0 f(x) dx$
Correctly written or $F(0) - F(-1)$ correct

B1

$[0] - \left[\frac{1}{2} + 1 - 5 \right]$

*Attempt to sub limit(s) of -1 (and 0)
 Must have integrated*

M1

Area under curve $3\frac{1}{2}$
CSO (no fudging)

A1

Area of shaded region = $3\frac{1}{2} - 2\frac{1}{2} = 1$

FT their integral and triangle (very generous)

B1 ✓

4

(b) (i) $\frac{dy}{dx} = 15x^4 + 2$

One term correct

M1

All correct (no +c etc)

A1

when $x = -1$, gradient = 17

CSO

A1

3

(ii) $y = \text{"their gradient"} (x + 1)$

*Must be finding **tangent** – not normal
any form e.g. $y = 17x + 17$*

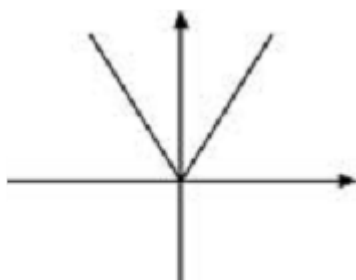
B1 ✓

1

[14]

Q21.

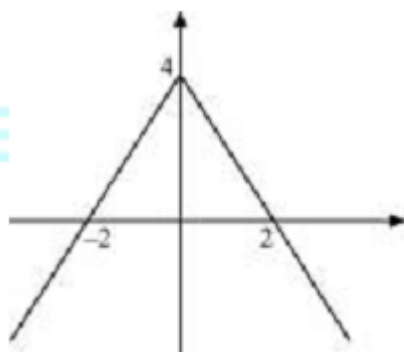
(a)



B1

1

(b)



Shape inverted V in all four quadrants

M1

Symmetrical about y-axis

A1

Coordinates

A1

3

(c) $4 - |2x| = x$

M1

$$4 - 2x = x \quad x = \frac{4}{3}$$

Attempt to solve

A1

$$4 + 2x = x \quad x = -4$$

And no others

A1

3

(d) $-4 < x < \frac{4}{3}$

Either correct

M1

Other solution and no extras

$$\text{SC} - 4 \leq x \leq \frac{4}{3} \quad \text{B1}$$

A1

2

[9]

Q22.

(a) (i) $p(2) = 8 + 4 - 20 + 8$

Finding p(2) M0 long division

M1

$$= 0, \Rightarrow x - 2 \text{ is a factor}$$

*Shown = 0 AND conclusion/ statement
about $x - 2$ being a factor*

A1

2

(ii) Attempt at quadratic factor
or factor theorem again for 2nd factor

M1

$$x^2 + 3x - 4$$

or $(x+4)$ or $(x-1)$ proved to be a factor

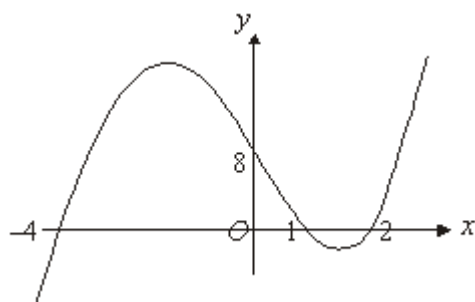
A1

$$p(x) = (x - 2)(x + 4)(x - 1)$$

A1

3

(b)



Graph through (0,8) 8 marked

B1

Ft "their factors" 3 roots marked on x-axis

B1ft

Cubic curve through their 3 points

M1

Correct including x-intercepts correct
Condone max on y-axis etc or slightly
wrong concavity at ends of graph

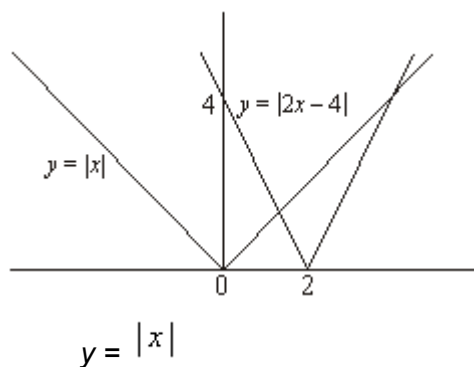
A1

4

[9]

Q23.

(a) (i) (ii)



B1

1

2 branches mod graph $x > 0$ for $y = 0$

for 2, 4

M1

A1

2

(b) (i) $x = 2x - 4, x = 4$

B1

$$-x = 2x - 4$$

M1

$$x = \frac{4}{3}$$

OE one value only

A1

3

Alternative:

$$x^2 = (2x - 4)^2$$

M1

$$x = 4, \frac{4}{3}$$

A1A1

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(ii) $\frac{4}{3} < x < 4$

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$\frac{4}{3}, 4$ (ft) identified as extremes

M1

CAO

A1

2

[8]

Q24.

(a) (i) $(5000 - x)$ seen in a product
 Could be implied, eg $5000a - xa$

B1

$$\frac{dx}{dt} = kx(5000 - x)$$

B1

2

(ii) $200 = k \times 1000 \times (5000 - 1000)$

$$\frac{dx}{dt} = 200, x = 1000 \quad \text{in their diff. equation}$$

Condone ts and $t = 0$ for M1

M1

$$k = 0.00005$$

CAO OE

A1

2

(b) (i) $t = 4 \ln \left(\frac{4 \times 2500}{5000 - 2500} \right) = 5.5 \text{ (hours)}$

$$x \rightarrow 2500 \text{ (or } 4 \ln 4 \text{)}$$

CAO

M1

A1

2

(ii)
$$e^{\frac{30}{4}} = \frac{4x}{5000 - x}$$

(ii)

OE

B1

M1

$$5000 \times e^{7.5} = x(4 + e^{7.5})$$

Soluble for x

m1

$$x = 4988.96.. \Rightarrow 4989 \text{ rabbits infected}$$

Or 4988 or 4990; integer value only

A1

4

[10]

Q25.

(a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 \, dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

When $t = 0$, $v = 20$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

only for fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9 \text{ or } 20 \ln 2$$

A1

3

(d) Work done by force is change in KE of car

M1

$$= \frac{1}{2}m(20)^2 - \frac{1}{2}m(10)^2$$

A1

$$= 150 \text{ m}$$

A1

3

[11]

Q26.

(a) Using $F = ma$

$$-0.05mv = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.05v$$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 \, dt$

B1

$$\ln v = -0.05t + c$$

$$v = Ce^{-0.05t}$$

Need first 2 terms

M1

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When $t = 0$, $v = 20$

$$\therefore C = 20$$

$$v = 20e^{-0.05t}$$

fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$$e^{0.05t} = 2$$

A1

$$\therefore t = \frac{1}{0.05} \ln 2$$

$$= 13.9$$

Accept 20 ln 2

A1

3

[8]

Q27.

- (a) No air resistance / Only gravity or weight

B1: Acceptable assumption

B1

1

- (b) (i) $0.2 \times 8 = 0.2 \times 9.8 - R$

M1: Three term equation of motion

A1: Correct equation

M1A1

$$R = 0.36 \text{ N}$$

A1: Correct magnitude of the resistance force

A1

3

- (b) (ii) **Increases** as the speed increases

B1: Correct explanation

B1

1

- (c) (i) $\pm 9.8 \text{ ms}^{-2}$

B1: CAO

B1

1

- (ii) **Decreases** towards zero

B1: Correct explanation

B1

1

[7]